

IC 272: Machine Learning

Course Code	IC 272
Course Name	Machine Learning
L-T-P-C	2-0-2-3
Prerequisites	IC 111 – Linear Algebra, IC 152 – Computing and Data Science, IC 252 - Probability and Statistics (Data Science 2)
Intended for	B.Tech. 2nd year students
Approval	28th Senate (Nov 2020), in the B.Tech Civil Engineering curriculum; 55th BoA (May 2024); 44th Senate (Sep 2024), in the B.Tech (all branches) - Institute Core curriculum
Revised	47th BoA (Sep 2022) and 37th Senate (Oct 2022) - renamed from “Data Science III”

Course Contents

- **Data preprocessing:** Data cleaning – missing values, noisy data; Data integration and transformation – normalization; Data reduction – dimension reduction and principal component analysis (PCA) [5 Lecture]
- **Introduction to machine learning:** Supervised and unsupervised learning [1 Lecture]
- **Supervised learning with applications in classification problems:** Bayes classifier with unimodal and multimodal density - maximum likelihood estimation, expectation-maximization (EM) algorithm (only at idea level), K-nearest neighbor methods, decision trees, neural networks [8 Lectures]
- **Supervised learning - regression:** Linear regression, polynomial regression, regression using neural networks [8 Lectures]
- **Unsupervised Learning Algorithms - Clustering:** K-means and fuzzy-K-means clustering, density based clustering (DBSCAN) [6 Lectures]

Laboratory Modules

Lab to be conducted on a 3-hour slot. It will be conducted in tandem with the theory course so the topics for problems given in the lab are already initiated in the theory class. The topics taught in the theory course should be appropriately be sequenced for synchronization with the laboratory.

- Lab1 : Data Preprocessing – data cleaning and normalization
- Lab2 : Dimension reduction and PCA
- Lab3 : Classification using Bayes classifier with unimodal density
- Lab4 : Classification using Bayes classifier with multimodal density
- Lab5 : Classification using K-nearest neighbor methods and decision trees
- Lab6: Classification using neural networks
- Lab7: Linear regression
- Lab8: Polynomial regression
- Lab9: Neural network based regression
- Lab10: Clustering using K-means and fuzzy-K-means
- Lab11: Clustering using DBSCAN

Text Books

1. C. M. Bishop, **Pattern Recognition and Machine Learning**, Springer, 2006.
2. C. Muller and S. Guido, **Introduction to Machine Learning with Python: A Guide for Data Scientists**, O'Reilly, 2017

References

1. J. Han and M. Kamber, **Data Mining: Concepts and Techniques**, 3rd Edition, Morgan Kaufmann Publishers, 2011
2. S. Theodoridis and K. Koutroumbas, **Pattern Recognition**, Academic Press, 2009.
3. T. Hastie, R. Tibshirani and J. Friedman, **The Elements of Statistical Learning: Data Mining, Inference, and Prediction**, 2nd Edition, 12th Reprint, Springer, 2017
4. R. O. Duda, P. E. Hart and D. G. Stork, **Pattern Classification**, John Wiley, 2001.